

POPULATION GROWTH OF *DAPHNIA MAGNA* UNDER MULTIPLE STRESS CONDITIONS: JOINT EFFECTS OF TEMPERATURE, FOOD, AND CADMIUM

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Abstract—Aquatic organisms in the field often are exposed to combinations of stress factors of various origins. Little is known of the interaction between different types of stressors; hence, the predictability of their joint effects is low. Therefore, the present study analyzed the joint effects of temperature, food, and cadmium on the population growth rate of the water flea *Daphnia magna*. The results revealed that temperature, food, and cadmium, as well as their interactions, were important factors that influenced life-history parameters and, as a consequence, the population growth rate of *D. magna*. In general, population growth rate increased at high temperature and food level but decreased when cadmium was present. The positive effect of temperature on population growth rate was smallest at limiting food levels. Negative effects of cadmium on the growth rate were enhanced at elevated temperatures, whereas high food levels protected the daphnids from adverse effects of cadmium. To avoid over- or underestimation regarding the toxicity of substances to field populations, results of standard toxicity tests should be applied in a location-specific way.

Keywords—Multiple stresses *Daphnia magna* Food Temperature Cadmium

INTRODUCTION

Aquatic organisms in natural waters often are subjected to various stress factors of physical, chemical, and biological origin. A substantial amount of information regarding the interacting effects of chemicals is available from mixture-toxicity studies [1,2], but how combinations of different types of stressors interact on the population level is still relatively unknown. The present study focused on the joint effects of two major natural stress factors, temperature and food, and of a chemical stressor, cadmium, on life-cycle parameters (i.e., survival, reproduction) of the water flea *Daphnia magna*. These chronic endpoints describe trends in population density or abundance and predict future population sizes [3], making them relevant tools in assessing the fate of field populations under multiple stress conditions. Temperature and food were chosen because these factors may be highly variable in the field and are of major importance to the physiological state of organisms. Cadmium was chosen as a model toxicant because it still causes problems in Dutch inland waters, even though its concentrations are declining [4].

Based on previous literature reviewed by Heugens et al. [5], an interaction between the three factors was expected. A temperature rise increases metabolic rate in ectothermic organisms [6] and, thus, feeding activity. For cladocerans, this activity also depends on food level, with feeding rate increas-

ing with food concentration up to a threshold concentration, the incipient limiting level, above which the feeding rate remains relatively constant [7]. Increased food uptake results in a higher amount of energy available for growth and reproduction. Hence, the population growth rate of animals living at high temperatures and with abundant food is expected to be higher than that of animals living at low temperatures and with poor food conditions. Cadmium, on the other hand, affects the population growth rate of cladocerans [8–10]. Acute experiments revealed that cadmium is more toxic to daphnids at high than at low temperature [11–13]. At high food levels, toxicity may decrease if the greater amount of energy can be directed to withstand toxic stress. Alternatively, the bioavailability of cadmium may change with food level, because the negatively charged surface of algal cells, a common food source for daphnids, has a high affinity for positively charged metal cations, resulting in a reversible binding of cadmium ions to the cell surface [14]. The hypotheses regarding the interaction between the three variables as stated above were tested by exposing *D. magna* to several cadmium concentrations in the water at combinations of three temperatures and three food levels, ranging from 10 to 26°C and from 0.50 to 2.0 mg C/L of a concentrated suspension of *Selenastrum capricornutum*, respectively. Tissue cadmium concentrations were measured to relate toxic effects caused by cadmium to the amount of toxicant accumulated by the daphnids. In this way, potential effects of temperature and algal density on the bioavailability of cadmium were determined as well.

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Table 1. Experimental setup with actual cadmium concentrations ($\mu\text{g/L}$) measured in the different temperature, food, and cadmium treatments^a

Food level (mg C/L)	Actual cadmium concentration ($\mu\text{g/L}$)			
	10°C	20°C	20°C	26°C
0.5	<DL	<DL	<DL	<DL
	48.6 (7.8)	14.4 (3.1)	27.4 (4.3)	19.7 (2.9)
	78.2 (9.2)	23.0 (3.6)	76.8 (5.8)	29.4 (5.1)
	136 (11)	37.8 (4.3)		49.1 (7.3)
1.0	<DL	<DL	<DL	<DL
	46.5 (8.7)	23.4 (4.3)		19.5 (3.2)
	78.0 (9.1)	37.3 (3.8)		29.2 (4.1)
	136 (11)	85.7 (4.9)		48.2 (6.5)
2.0	<DL	<DL	<DL	<DL
	132 (12)	85.0 (5.2)		30.2 (3.8)
	228 (15)	140 (6.4)		50.5 (6.7)
	358 (20)	234 (9.0)		80.8 (7.4)

^a Values in parentheses denote standard errors. <DL = below detection limit (control treatment). *Selenastrum capricornutum* was used as a food source.

MATERIALS AND METHODS

Culture conditions

Daphnia magna cultured in our laboratory descended from a culture kept at the Institute for Inland Water Management and Waste Water Treatment (RIZA, Lelystad, The Netherlands). The culture consisted of cohorts with a density of 20 daphnids per liter of artificial Elendt M7 medium [15]. A cohort was kept for four to five weeks, after which a new cohort was started with at least third-brood neonates (age, <24 h). Three times per week, the medium was renewed, and juveniles were removed. The culture was maintained under a 16:8-h light:dark photoperiod and at a temperature of 20°C. On working days, the daphnids were fed with 2 mg C/L of a concentrated suspension of *S. capricornutum*. The algae were cultured in a chemostat in Woods Hole medium [16]. Every week, algae were collected and centrifuged at 3,000 rpm for 10 min. After removal of the supernatant, the algae were resuspended in Elendt M7 medium, and the total organic carbon concentration of the suspension was measured with a total carbon analyzer (model 700; OI, College Station, TX, USA). The suspension was stored at 4°C in a dark room until used for feeding.

Chronic toxicity tests

The standard *Daphnia* reproduction test [15] was adjusted to study the influence of temperature, food, and cadmium on life-history traits of *D. magna*. Each experiment consisted of 12 treatments, which were composed of combinations of four cadmium concentrations (including one control) and three food levels (0.50, 1.0, and 2.0 mg C/L). The experiments were consecutively performed at 10, 20, and 26°C. In the 20°C experiment, the cadmium concentrations in the low-food treatment proved to be too low to exert effects. Therefore, an additional experiment with higher exposure concentrations was performed at 20°C. The experimental design is summarized in Table 1. The temperatures used are within the range of temperatures observed in the field. Food levels ranged from concentrations used in the standard *Daphnia* reproduction test [15] to lower food levels that are common in aquatic ecosystems. The lowest food level was derived from pilot experiments in batch systems and proved to sustain sufficient reproductive

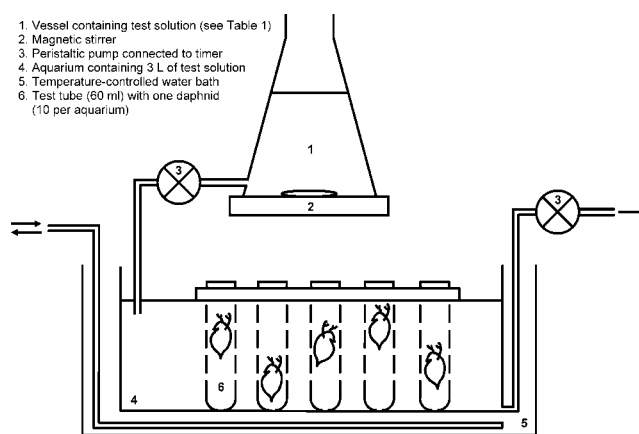


Fig. 1. Intermittent flow system.

output to be able to analyze negative effects resulting from cadmium exposure. This low food level is close to the average biomass of green algae in the eutrophic Lake IJsselmeer in The Netherlands, for which a food level of 0.375 mg C/L has been reported [17].

To maintain constant cadmium concentrations in the experiments and to avoid loss of algae from the water column by sedimentation, an intermittent flow-through system was used (Fig. 1). By the use of this flow-through system, the algal concentration in the water column was kept relatively constant, and distortion of the results by food shortage of the daphnids at elevated temperatures because of higher filtration rates was prevented. Per cadmium and food treatment in each temperature experiment, 10 neonates (age, <24 h) of at least the third brood were placed individually in polypropylene tubes, from which the middle part had been replaced by gauze. The 10 tubes were positioned together in a plastic aquarium that had been filled previously with 3 L of test solution. The test solutions consisted of artificial Elendt M7 medium [15] with different cadmium and food levels (Table 1) that were prepared in 3-L Erlenmeyer flasks and stirred for at least half an hour before usage. This duration was considered to be sufficient to reach equilibrium between cadmium in solution and cadmium adsorbed to algal cell surfaces [18]. Eight times a day, 375 mL of the corresponding solution was discharged into each aquarium by peristaltic pumps connected to a timer, and the same volume was removed from the aquarium simultaneously. In this way, a volume of test solution equivalent to the volume in the aquarium flowed through the system daily. A 16:8-h light:dark photoperiod was used. The temperature was kept constant by placing the aquaria in temperature-controlled water baths.

To determine actual cadmium concentrations in the water, 1.5-mL water samples were taken just before and 1 h after the addition of freshly made solution and centrifuged at 3,000 rpm for 10 min. One milliliter of supernatant was transferred to a new Eppendorf container and acidified with 20 μL of 70% nitric acid (Ultrex®, J.T. Baker, Phillipsburg, NJ, USA), after which they were analyzed by air-acetylene flame atomic absorption spectrometry (model 1100B; Perkin-Elmer, Norwalk, CT, USA). Stability of the algal concentrations was checked by measuring cell numbers and volumes in 5-mL water samples, diluted with 15 mL of salt solution (1.2% NaNO_3), with a Coulter counter. When the food level in a treatment started to increase because of sedimentation of the algae, the gauze tubes, with the daphnids in the remaining water at the bottom of the

tube, were removed from the aquarium. The test medium was transferred to an Erlenmeyer flask, and the bottom of the aquarium was flushed with double-distilled water to remove sedimented algae, after which the test medium and the test tubes were put back in the aquarium. During an experiment, this occurred, at most, once per treatment. Because temperature accelerates physiological processes, a certain exposure time at high temperature represents a longer period of physiological time compared with the same exposure time at low temperature. To prevent problems with interpretation, the test duration was based on physiological time (i.e., the time in which the daphnids produced three clutches of neonates). This physiological endpoint was chosen because population growth rate approached its maximum value after production of the third brood (see below). This resulted in test periods of 41, 15, and 11 d at 10, 20, and 26°C, respectively.

Each day, the following life-history traits of the daphnids were recorded: Survival, presence of eggs in the brood pouch, and number of living neonates produced. Newborn neonates were removed. From these life-history characteristics observed each day, the age at maturity (i.e., the day on which the first clutch of eggs was deposited in the brood pouch) and the age at first reproduction were determined. The life-history parameters were integrated by calculating the population growth rate according to the Euler-Lotka equation:

$$\sum_{x=0}^{x=x_m} l_x m_x e^{-rx} \quad (1)$$

where l_x is the fraction surviving until age x , m_x is the number of offspring produced per surviving female between age x and $x + 1$, r (/d) is the population growth rate, and x_m is the maximum age. The outcome of Equation 1 was obtained by iterative calculations, resulting in only one value for r per treatment. Because no uncertainty could be calculated from this single value, the jackknife method was used to generate a variance, as described previously by Meyer et al. [19].

To determine somatic growth of the daphnids during the test period, body lengths (measured from the middle of the eye to the base of the tail) of neonates (age, <24 h) descending from the same cohort as the test animals were measured at the start of the experiments using a Leica 1 MZ 8 Microscope equipped with a Leica DC100 Digital Camera (Leica Geosystems Products, Rijswijk, The Netherlands) using the computer program Research Assistant 3 (RVC, Hilversum, The Netherlands). This was repeated at the end of the experiment for surviving daphnids. For each treatment, growth was calculated as the difference in average body length at the start and at the end of the experiment. After length measurement, the animals were kept in clean, double-distilled water for 10 min, after which one to three daphnids were pooled in 2-ml polyethylene tubes. Following the microdestruction method as described previously by Timmermans et al. [20], the animals were lyophilized, weighed, and digested in 65% nitric acid (Ultrex, J.T. Baker) and 30% hydrogen peroxide (Fluka, Buchs, Swiss). The concentrated samples were diluted with 500 μ l of acidified analytical grade water (5 ml of 70% nitric acid per liter; Ultrex, J.T. Baker) and analyzed by air-acetylene flame or graphite furnace atomic absorption spectrometry (model 5100PC/HGA600/AS60 equipped with Zeeman background correction; Perkin-Elmer), depending on the metal concentration in the samples. To determine the maternal investment per neonate (as average neonate size) and the investment in total reproductive output (as average neonate size $\cdot$ brood size), body

lengths of neonates descending from the third brood were measured as well.

Statistical analyses

Data for all parameters were tested for homogeneity and normality. When these conditions were met, a three-way analysis of variance (ANOVA) was performed. Otherwise, a three-way ANOVA was performed on rank data. Rank transformation was chosen because the variance of rank data is automatically stable [21]. Cadmium treatments were compared with the corresponding control treatment by t tests using Bonferroni adjustment with an overall error of 0.05.

As mentioned above, the test period was based on a physiological age, resulting in a longer exposure time at low temperature than at high temperature. Because of this difference in test duration, life span was analyzed within temperature groups by performing a two-way ANOVA on rank data instead of a three-way ANOVA. The life span of animals that died within the first 24 h was set at 0 d, whereas the life span of an animal that lived until the end of the experiment equaled the test duration. This approach may have led to an underestimation of the effects on survival, because the differences in life span between treatments in which daphnids died during the experiment and those in which no mortality occurred were smaller than if the test duration would have been prolonged.

RESULTS

Cadmium accumulation

Tissue cadmium concentrations were determined in daphnids that survived until the end of the experiment (Fig. 2). Data regarding only one exposure concentration were available for the 10°C and high-food treatments, because no daphnids survived the other cadmium treatments. The corresponding control treatment was lost during the experiment as well, and the experiment had to be continued without the control group. Nevertheless, sufficient treatments remained to determine the effects of cadmium and food level and their interactions on *D. magna* at this temperature.

The cadmium concentration in the daphnids significantly increased with higher cadmium concentrations in the water, higher temperatures, and lower food levels. These effects were magnified when a combination of these conditions was applied, as indicated by the significant interactions between temperature and cadmium and between food and cadmium (three-way ANOVA) (Table 2).

Survival

Survival of the daphnids during the experiments is given in Figure 3. From these survival curves, average life span was calculated for all exposure conditions (not shown). The effects of cadmium and food level on life span were analyzed within each temperature treatment using two-way ANOVA (Table 2). At 10°C, cadmium significantly decreased life span, whereas no effect of food was found. At 20°C, both cadmium and food had a significant effect on survival. Cadmium decreased life span, but the influence of food level on life span was not consistent, with no clear trend being seen. At 26°C, only the food concentration significantly influenced life span; again, no clear pattern could be observed. Although the life span of the daphnids at the highest cadmium concentration was significantly shorter than in the control (t tests with Bonferroni correction), the high variation in life span of the daphnids in the cadmium treatment was the reason that cadmium was a non-

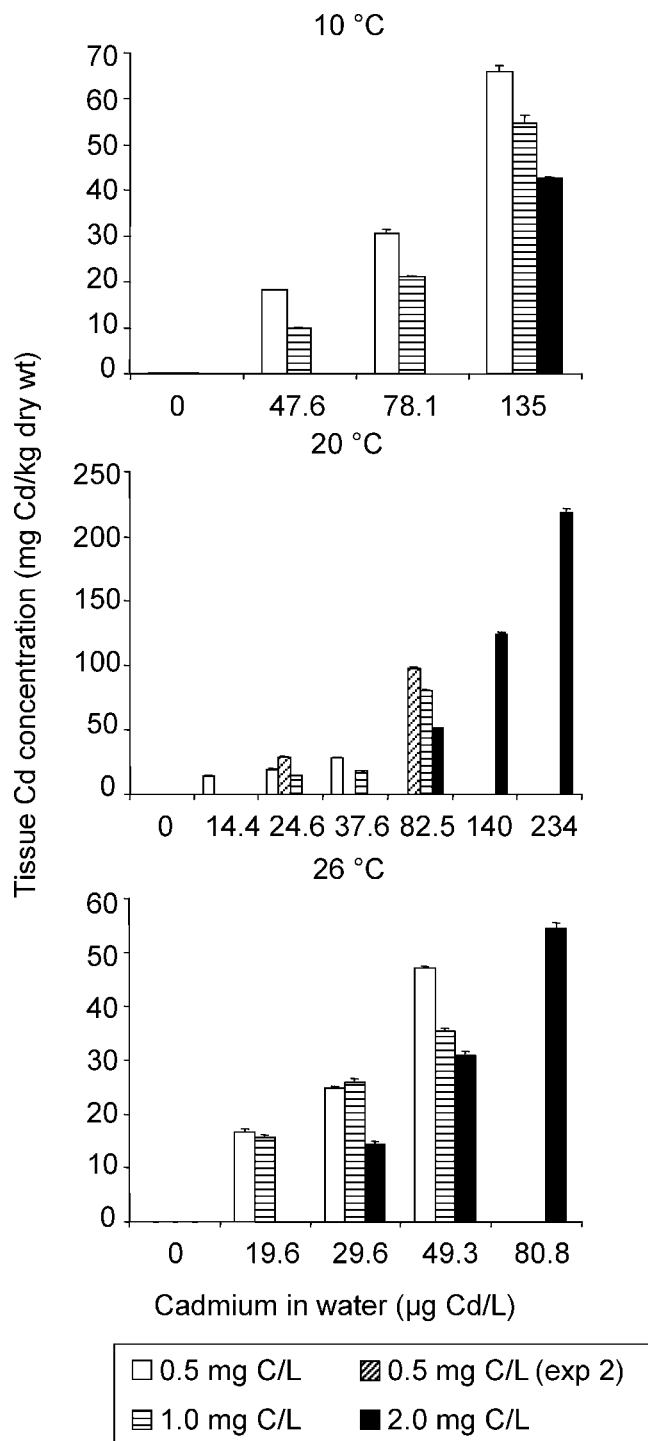


Fig. 2. Tissue cadmium concentrations of surviving *Daphnia magna* at 10, 20, and 26°C and 0.5 (experiment 1, white bars; experiment 2, diagonally shaded bars), 1.0 (horizontally shaded bars), and 2.0 mg C/L (black bars) of a concentrated suspension of *Selenastrum capricornutum* as a function of the cadmium concentration in the water. Error bars represent standard errors.

significant factor with regard to life span. No interactions between cadmium and food were present at any of the tested temperatures.

Growth

The effects of temperature, food, and cadmium, as well as their two- and three-factor interactions on growth, were all

significant (three-way ANOVA) (Table 2). Figure 4 shows the increase in body length during the test period for daphnids that were still alive at the end of the experiment. The daphnids reached a larger body size at low temperature and high food level. Exposure to cadmium decreased growth, and this effect was enhanced by high temperature and low food conditions.

Reproduction

Age at maturity and first reproduction. In Figure 5, wherein the cumulative number of neonates per living female as a function of age of the females is given, the age at first reproduction can be seen. Results of a three-way ANOVA (Table 2) indicated that elevated temperature and food level significantly reduced the age at maturity and the age at first reproduction of the daphnids, whereas exposure to cadmium postponed the onset of reproduction. The significant two- and three-factor interactions point out that the effects of cadmium on the age at maturity were enhanced by high temperature and low food level. With respect to the age at first reproduction, only the interaction between food and cadmium was significant, which implied that a high food level protected the daphnids against adverse effects of cadmium.

Total number of neonates. The total number of neonates produced was significantly affected by temperature, food, and cadmium (three-way ANOVA) (Table 2). Except for the interaction between temperature and food, all two- and three-factor interactions were significant as well. Figure 5 shows that the number of neonates born during the test period was significantly higher at 10°C than at 20 and 26°C, at which the numbers were similar. High food levels increased the reproductive output. When comparing effects of cadmium at the different food and temperature treatments, it appeared that cadmium affected reproductive output more at low food levels and high temperatures.

Maternal investment in reproduction. Although the effects of temperature, food, and cadmium, as well as the joint actions of temperature and food and of temperature and cadmium, on maternal investment per neonate (as neonate size) were significant (three-way ANOVA) (Table 2), no clear relationship between the three factors and neonate size were found. For this reason, no figures are shown for this parameter. However, distinct relationships were found for the total investment in reproduction (as neonate size·brood size) (Fig. 6). The total investment was affected by temperature, food, and cadmium. Significant interactions between temperature and cadmium, and between temperature, food, and cadmium, were present as well (three-way ANOVA) (Table 2). In general, the overall investment in reproduction increased with low temperature and high food level and decreased when the animals were exposed to cadmium. The negative effects of cadmium were increased at elevated temperature. Because no effects were observed for any of the three factors on neonate size, the differences in total investment could be attributed entirely to the differences in brood sizes.

Population growth rate

Population growth rate was significantly affected by temperature, food, and cadmium, as well as by their two-factor interactions (three-way ANOVA) (Table 2). Figure 7 shows the population growth rate at the different exposure regimes. Population growth rate increased with elevated temperature and food levels and decreased with increasing concentration of cadmium. However, at 26°C and medium food level, cad-

Table 2. *p* Values for two- and three-way analysis of variance for effects of temperature (T), food level (F), and cadmium (Cd) and their interactions on life span, growth, age at first reproduction, total number of neonates produced per mother, and population growth rate of *Daphnia magna*^a

Effect parameter	T	F	Cd	T × F	T × Cd	F × Cd	T × F × Cd
Tissue cadmium concentration	<0.01	<0.01	<0.01	0.75	<0.01	<0.01	0.08
Life span							
10°C		0.27	<0.01			0.06	
20°C		0.01	0.02			0.59	
26°C		0.04	0.62			0.09	
Somatic growth							
Age at maturity	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Age at first reproduction	<0.01	<0.01	<0.01	0.93	0.07	<0.01	0.36
Total number of neonates	<0.01	<0.01	<0.01	0.06	<0.01	<0.01	0.01
Population growth rate	<0.01	<0.01	<0.01	0.02	<0.01	<0.01	0.10
Investment in individual neonates	<0.01	<0.01	<0.01	0.04	<0.01	0.15	0.58
Investment in total reproductive output	<0.01	<0.01	<0.01	0.61	<0.01	0.06	0.04

^a Except for growth data, all data were rank transformed before analysis.

mium concentrations up to 29.6 µg/L caused an increase in population growth rate relative to the control treatment. At 10°C, population growth rate was hardly influenced by food level, but at elevated temperatures of 20 and 26°C, high food levels were observed to stimulate population growth. In general, cadmium became more damaging at low food levels. The cadmium-induced decrease in growth rate was amplified at elevated temperature, because similar exposure concentrations resulted in a larger reduction in growth rate than at low temperature.

DISCUSSION

Organisms living under field conditions are subjected to a variety of environmental stressors. Still, relatively little is known about the combined effects of these stress factors. Although a substantial amount of literature is available concerning the distinct effects of temperature, food, and cadmium, less information is at hand concerning the joint effects of two fac-

tors. Studies involving three-factor interactions are exceptional.

The results of the present study revealed that temperature, food, and cadmium, as well as their interactions, were important factors that influenced life-history parameters and, as a consequence, population growth rate of *D. magna*.

The age at maturity and the age at first reproduction decreased at high temperature, enhancing population growth. This is in accordance with the common observation that higher temperatures accelerate physiological processes in ectotherms, which most aquatic organisms are [6]. In contrast, somatic growth of the daphnids and number of neonates produced were higher at 10°C than at 20 and 26°C. Possibly, the increased metabolic costs at elevated temperatures resulted in less energy available for growth. Because smaller daphnids produce less offspring [22], the lower reproduction at elevated temperatures could be caused by the smaller body size of the daphnids at these temperatures. However, this decrease in reproduction did

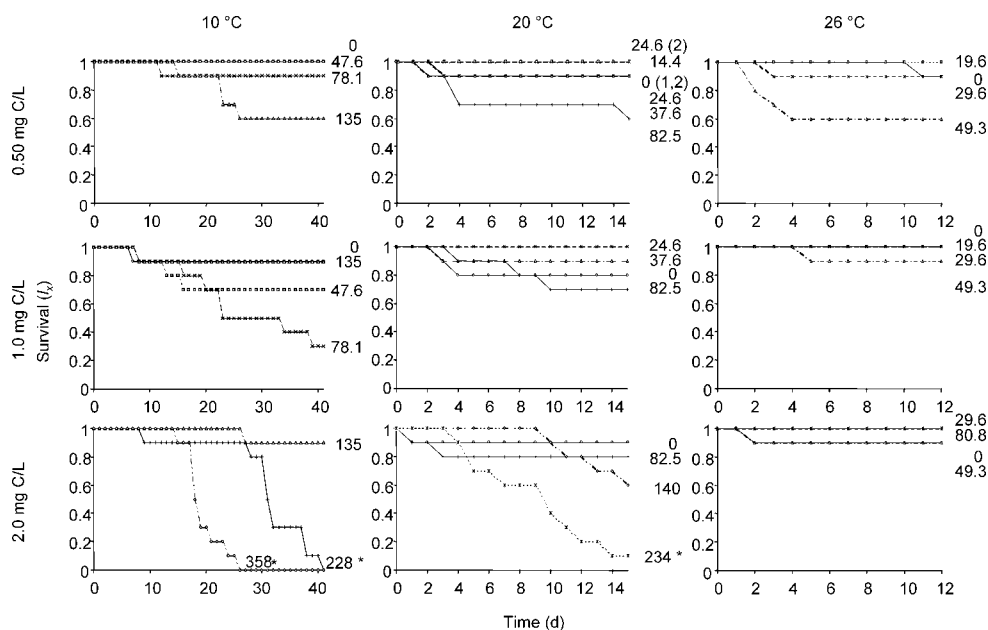


Fig. 3. Survival of *Daphnia magna* exposed to various cadmium concentrations (as indicated by the numbers on the right side of the graphs) at 10, 20, and 26°C and 0.5 (experiments 1 and 2), 1.0, and 2.0 mg C/L of a concentrated suspension of *Selenastrum capricornutum* as a function of age. An asterisk indicates a significant difference in life span relative to the corresponding control (or lowest cadmium treatment for the 10°C and high-food treatment).

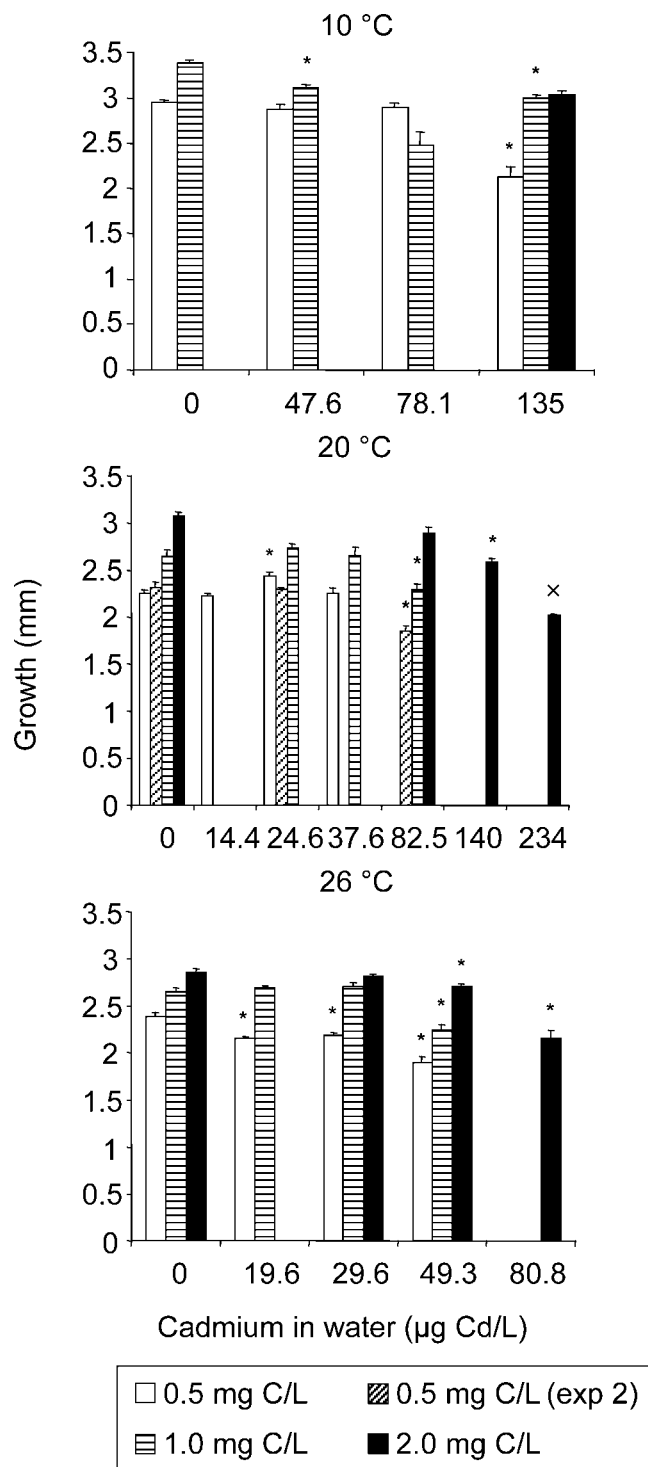


Fig. 4. Growth of surviving *Daphnia magna* at 10, 20, and 26°C and 0.5 (experiment 1, white bars; experiment 2, diagonally shaded bars), 1.0 (horizontally shaded bars), and 2.0 mg C/L of a concentrated suspension of *Selenastrum capricornutum*, as a function of the cadmium concentration in the water. Error bars represent standard errors. An asterisk indicates a significant difference in growth relative to the corresponding control (or lowest cadmium treatment for the 10°C and high-food treatment), and a × symbol indicates that no statistical comparison with the corresponding control could be made because too few data were available.

not result in a lower population growth rate at 20 and 26°C compared with that at 10°C. Similar outcomes were reported for *D. magna* by Sakwińska [23]. Daphnids raised at lower temperature matured later and at a larger size, resulting in a higher number of offspring in the first two clutches, but population growth rate was still higher at elevated temperature.

Higher population growth rates were obtained at high food levels. This was due to a decrease in the age at which the daphnids started to reproduce in combination with an increase in the amount of neonates produced. The latter could be attributed to the larger body size of the daphnids when food was abundant. At high food levels, more energy is available for growth and reproduction [24], which is reflected by the better performance of well-fed relative to poorly fed daphnids.

Cadmium exposure decreased the population growth rate, because the metal affected all parameters except life span at 26°C. This is in accordance with the findings of previous studies [8–10].

The interaction between temperature and food significantly affected the age at maturity, growth, and population growth rate of the daphnids but none of the other life-history parameters studied. The increase in growth rate when temperature rose from 10 to 20°C was consistent for the three food levels, but at 26°C, the increase was smaller at low food levels relative to that at the higher food levels. Apparently, food deficiency in the low-food treatment increased with temperature. Because the algal concentration in the water column was relatively constant (data not shown), this deficiency was not a result of the test design but, more likely, of physiological processes, such as a higher metabolic rate. The findings suggest that at the lowest food concentration, food intake of the daphnids could no longer support the higher metabolic energy needs induced by high temperature, which resulted in a lower population growth rate. Indeed, many authors state that temperature effects are most pronounced at nonlimiting food conditions [25,26]. Clearly, the daphnids can gain from elevated temperature only when enough food is available to cope with the increased metabolic costs.

The two-factor interactions between temperature and cadmium and between food and cadmium influenced the population growth rate. This was caused by differences in the number of neonates produced under the tested conditions. Temperature and cadmium interacted in a way that temperature enhanced the negative effects of cadmium on the population growth rate. Previous studies also have reported increased cadmium toxicity at elevated temperatures in acute experiments [11–13], which was caused by enhanced cadmium uptake rates as well as by elevated intrinsic sensitivity of the daphnids [13]. Indeed, the tissue cadmium concentrations measured in the present study revealed that daphnids accumulated a higher amount of cadmium at elevated temperatures. Results of other chronic experiments were in accordance with the findings of the present study. For instance, decreased population growth rates at rising temperature were found for *Daphnia pulex* exposed to toxic cyanobacteria [27]. A similar temperature response was presented for survival of *Daphnia ambigua* exposed to *Chaoborus* sp. kairomone, but no effects on growth and reproduction were observed [28]. In both studies, the observed interactions between the stress factors were explained by increased metabolic costs associated with high temperatures. When an additional stress factor was present, the daphnids could no longer cope with these increased metabolic needs, either because the stress factor inhibited food intake

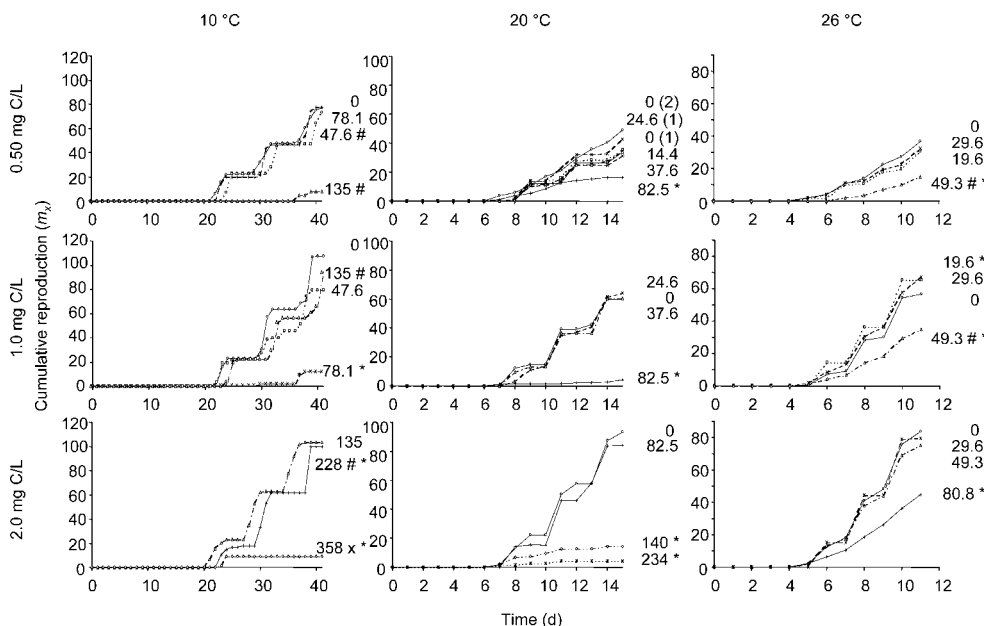


Fig. 5. Cumulative number of neonates produced by *Daphnia magna* exposed to various cadmium concentrations (as indicated by the numbers on the right side of the graphs) at 10, 20, and 26°C and 0.5 (experiments 1 and 2), 1.0, and 2.0 mg C/L of a concentrated suspension of *Selenastrum capricornutum* as a function of age. A # symbol and an asterisk indicate a significant difference in age at first reproduction and total number of neonates, respectively, relative to the corresponding control (or lowest cadmium treatment for the 10°C and high-food treatment), and a × symbol indicates that no statistical comparison with the corresponding control could be made because too few data were available.

(cyanobacteria) or because it induced a defense mechanism (kairomone).

The interaction between food level and cadmium became visible from the lower toxicity of cadmium in well-fed daphnids compared to poorly fed ones. Apparently, the high food level protected the daphnids against adverse effects of cadmium. This resulted in an earlier start of reproduction, a larger body size, and the production of a higher number of neonates, which led to higher population growth rates at the high food level than at the low food level. Studies that have focused on the influence of food level on chronic cadmium toxicity to the cladocerans *Daphnia carinata* and *Echinisca triserialis* also found increased effects of cadmium on population parameters with decreased food levels [29,30]. Cadmium disturbed the feeding processes, and it was suggested that food intake by poorly fed daphnids was insufficient to maintain normal functioning, especially when metabolic costs increased because of cadmium stress. Although Klüttgen and Ratte [31] found a similar relationship between cadmium and food for effects on juvenile development, they observed enhanced adverse effects of cadmium on growth and reproduction in adult daphnids held at high food concentrations. With respect to population growth rate, they reported that this rate was largely independent of food level. The authors suggested that the response of adults could be caused by the low calcium concentration in the test medium, which caused an increase in cadmium uptake via calcium-specific carriers, especially at higher food levels, as a result of an elevated metabolic rate. Although Klüttgen and Ratte [31] did not specify the calcium content, the total water hardness in their test water was fourfold lower than the calcium concentration in our test medium. This could explain the different interaction of cadmium and food observed in the present study.

Another possible mechanism is related to the binding of the metal onto algal cell surfaces. Tissue cadmium concentrations determined in daphnids that survived until the end of the

experiment revealed that less cadmium was accumulated by the daphnids held at high food concentration than in those held at low food concentration. Apparently, more cadmium ions were adsorbed per algal cell when food was less abundant. Allen et al. [14] found that cations such as free cadmium induced feeding inhibition at sublethal concentrations, in contrast to anionic toxicants that caused inhibition of feeding only at lethal concentrations. Reduced feeding rates in cadmium-exposed *D. magna* also were observed in other studies [18,29,30,32,33]. These findings indicate that adsorption of cadmium ions to the negatively charged surface of algal cells is a potential exposure route of cadmium to particle-grazing daphnids. When food uptake also was impaired in the present experiment, the effects would be higher at low food level, both because of the higher amount of cadmium adsorbed to the algae and because of the net food intake being affected more at low food levels. The effect on feeding is likely to affect growth and reproduction, finally resulting in reduced population growth rates.

Although a three-factor interaction between temperature, food, and cadmium was found for effects on age at maturity, growth, and the number of neonates, such an interaction was not observed for population growth rate. We know of only one other study that has dealt with chronic experiments using *Daphnia* organisms exposed to three stress factors: Folt et al. [34] analyzed the joint effects of temperature, food, and the detergent sodium dodecyl sulfate (SDS) on cumulative reproduction and survival of *Daphnia* organisms. Cumulative reproduction in multiple-stressor treatments generally was less than or equal to reproduction in single-stressor treatments. Comparable to our findings, reproduction rose with increasing temperature, until temperature became lethal. At low food level, reproduction started later and was lower than that at high food level. Despite these significant effects of temperature and food, no interaction between the two factors was found. The interaction between food and SDS was opposite to our findings:

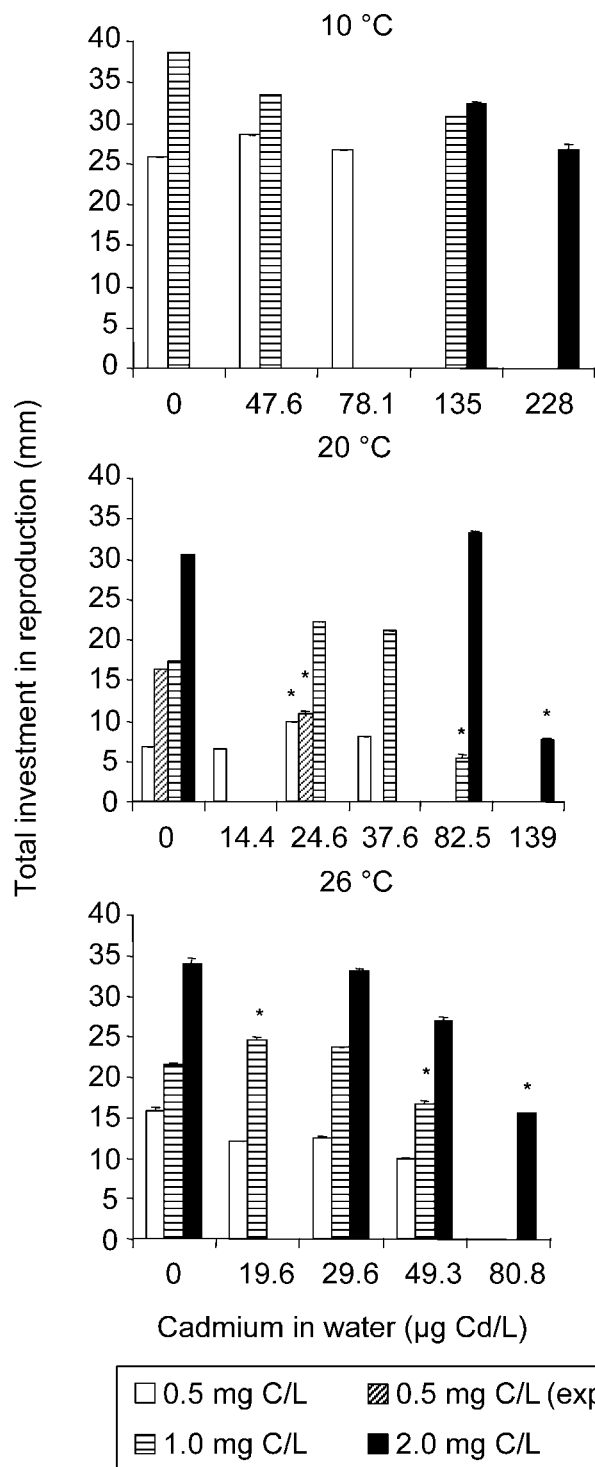


Fig. 6. Total investment in reproduction of *Daphnia magna* at 10, 20, and 26°C and 0.5 (experiment 1, white bars; experiment 2, diagonally shaded bars), 1.0 (horizontally shaded bars), and 2.0 mg C/L (black bars) of a concentrated suspension of *Selenastrum capricornutum* as a function of the cadmium concentration in the water. Error bars represent standard errors. An asterisk indicates a significant difference in total investment in reproduction relative to the corresponding control (or lowest cadmium treatment for the 10°C and high-food treatment).

SDS reduced reproduction to a greater extent at high food level than at low food level. A three-factor interaction revealed that the influence of food on toxicity was stronger at high temperatures. Similarly, toxicant-induced mortality increased

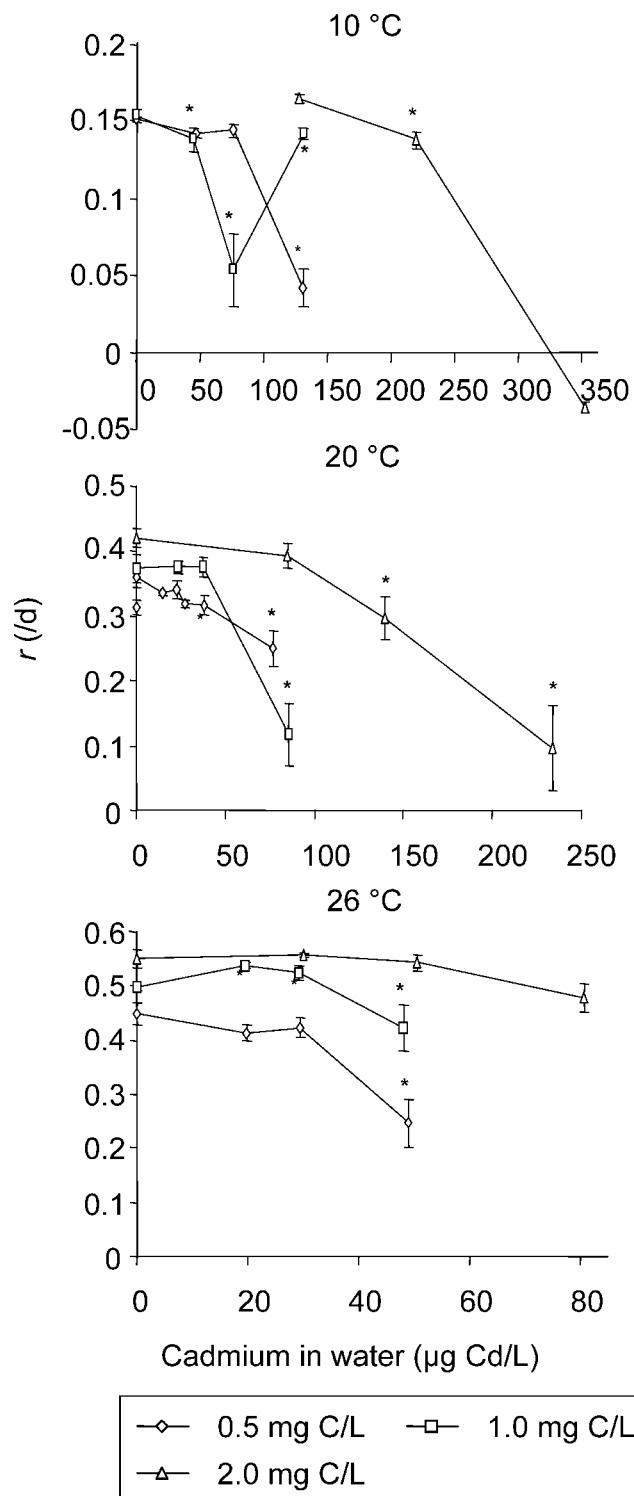


Fig. 7. Population growth rate of *Daphnia magna* at 10, 20, and 26°C and 0.5 (squares), 1.0 (diamonds), and 2.0 mg C/L (triangles) of a concentrated suspension of *Selenastrum capricornutum* as a function of cadmium concentration in the water. Error bars represent 95% confidence intervals. An asterisk indicates a significant difference in population growth rate relative to the corresponding control (or lowest cadmium treatment for the 10°C and high-food treatment).

at high temperature, and this effect was more pronounced at high food level. It was suggested that these interactions were related to increased toxicant uptake rates from the water and food at higher temperature and food concentration because of

elevated feeding rates. The difference in toxicant mode of action (SDS vs cadmium) could account for the contrasting results of Folt et al. [34] and the present study.

The present study is one of the few laboratory studies that experimentally specify the variability of toxicant-induced effects on population dynamics because of variable food levels and temperatures. It generally is accepted that wide margins of uncertainty are associated with quantifying toxic effects, and even greater uncertainties are introduced on the extrapolation of laboratory observations to populations in the field. The present study shows that a substantial part of this uncertainty can be explained by natural factors that modify toxicity through interactive effects at the level of individuals, ultimately showing up as effects on population dynamics.

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